1. INTERPRET DRAWING IAW STANDARDS PRESCRIBED BY ASME Y14.100.
2. UNLESS OTHERWISE SPECIFIED:
   ALL RESISTORS ARE 1%, 0.1W, 0603
   ALL CAPACITORS ARE 10%, 16V, 0603
3. PARTIAL REFERENCE DESIGNATIONS ARE SHOWN FOR COMPLETE.
4. SEE TABLE A FOR POWER AND GROUND INPUTS TO MICROCIRCUITS.
5. INTERPRET REFERENCE DESIGNATIONS AS FOLLOWS:
   EXAMPLE: U1 - 2
   COMPONENT PIN NUMBER
6. GENERIC PART NUMBERS ARE SHOWN FOR REFERENCE ONLY.
7. CONNECT CHASSIS TO THE MOUNTING HOLES / CHASSIS GROUND.
8. LATEST REVISION CHANGES: NaN
9. UNUSED REFERENCES: R40, R41, R42, C28, C29, C30, C38, C39, C40
   → DO NOT INSTALL THESE COMPONENTS. LEAVE THEM UNPOPULATED.

REVISIONS

ZONE	LTR	DESCRIPTION	DATE	APPR.
	-	RELEASED PER ENGINEERING CSULB-DEESC	260207	T STOLPE

TABLE A

REFERENCE DESIGNATION	POWER INPUT TO PINS							
	PWR	SRC						
	+24 V	6S BATT						
	+10 V	PSU						
	+3.3 V	BUCK						
	GND	RTN						

INSTALLATION GUIDE:

1. ESC BOARD GOES BELOW THE MCU BOARD. INSTALL NORMAL 1X12 FEMALE HEADER PINS INTO P2 AND P3.
2. A SWITCH AND APPROPRIATE FUSE SHALL BE SOLDERED TO THE BATTERIES POSITIVE WIRE. THE SWITCH ACTS AS MASTER POWER. REFER TO LITTLEFUSE FUSE FACTS AND FUSE SELECTION GUIDE TO DETERMINE THE APPROPRIATE FUSE.
3. AFTER FUSE AND SWITCH IS INSTALLED, SOLDER THE BATTERY/POWER SOURCE WIRES TO THE PADS LABELED BATT+ AND BATT-. BE CAREFUL TO NOT MIX THESE UP AS THERE IS NO REVERSE POLARITY PROTECTION.
4. C1 AND C2 SHALL BE INSTALLED ONTO THE BOTTOM LAYER. THE 3D HOUSING FOR THE ESC WILL BE DESIGNED TO HIDE THESE CAPS.
5. INSTALL 1X4 HEADER PINS INTO P4. BE CAREFUL TO INSTALL HEADER PINS WITH OR WITHOUT EXTENDED PINS DEPENDING ON THE STACK CONFIGURATION. THE ESC BOARD IS DESIGNED TO GO BELOW THE MCU BOARD.
6. MOUNTING HOLES ARE SIZED FOR M3 SCREWS. MOUNTING HOLES ARE NOT PLATED WITH COPPER.
7. THIS BOARD SUPPORTS AN EXTERNAL BUCK AT P1. REFER TO P1 ON THE BOARD FOR THE CORRECT PINS FOR THIS INSTALLATION.
8. DNI R1,R2,R3 IF THE DRV8323 IS CONFIGURED FOR SINGLE (1X) PWM MODE.
   FOR MCU BOARD ONLY: DNI R14, R15, R16, C22, C23, C24 IF THE DRV (6X) PWM MODE.
   DO NOT USE EXTERNAL BUCK WITHOOUT DISABLING DRV8323 BUILT-IN BUCK CONVERTER.
   REMOVE R12 IN THE CASE THAT A EXTERNAL BUCK CONVERTER IS USED.
   DNI C17 IF THE DRV8323 AMPLIFIERS ARE CONFIGURED FOR MOSFET VDS CURRENT SENSING.
   DNI R9, R10, R11 IF THE DRV8323 AMPLIFIERS ARE CONFIGURED FOR DC BUS CURRENT SENSING.
   DNI R27, R29, R32 IF THE DRV8323 IS CONFIGURED FOR SINGLE (1X) PWM MODE.
9. THE SOLDERABLE PADS FOR THE MOTOR ARE LABELED WITH THEIR RESPECTIVE PHASES (A, B, AND C).
10. IF MOUNTING TO A 3D-PRINTED HOUSING, TAKE INTO CONSIDERATION THE COMPONENTS INSTALLED ONTO THE BOTTOM LAYER. USE SCREW SPACERS/STANDOFFS TO NOT DAMAGE THESE COMPONENTS.

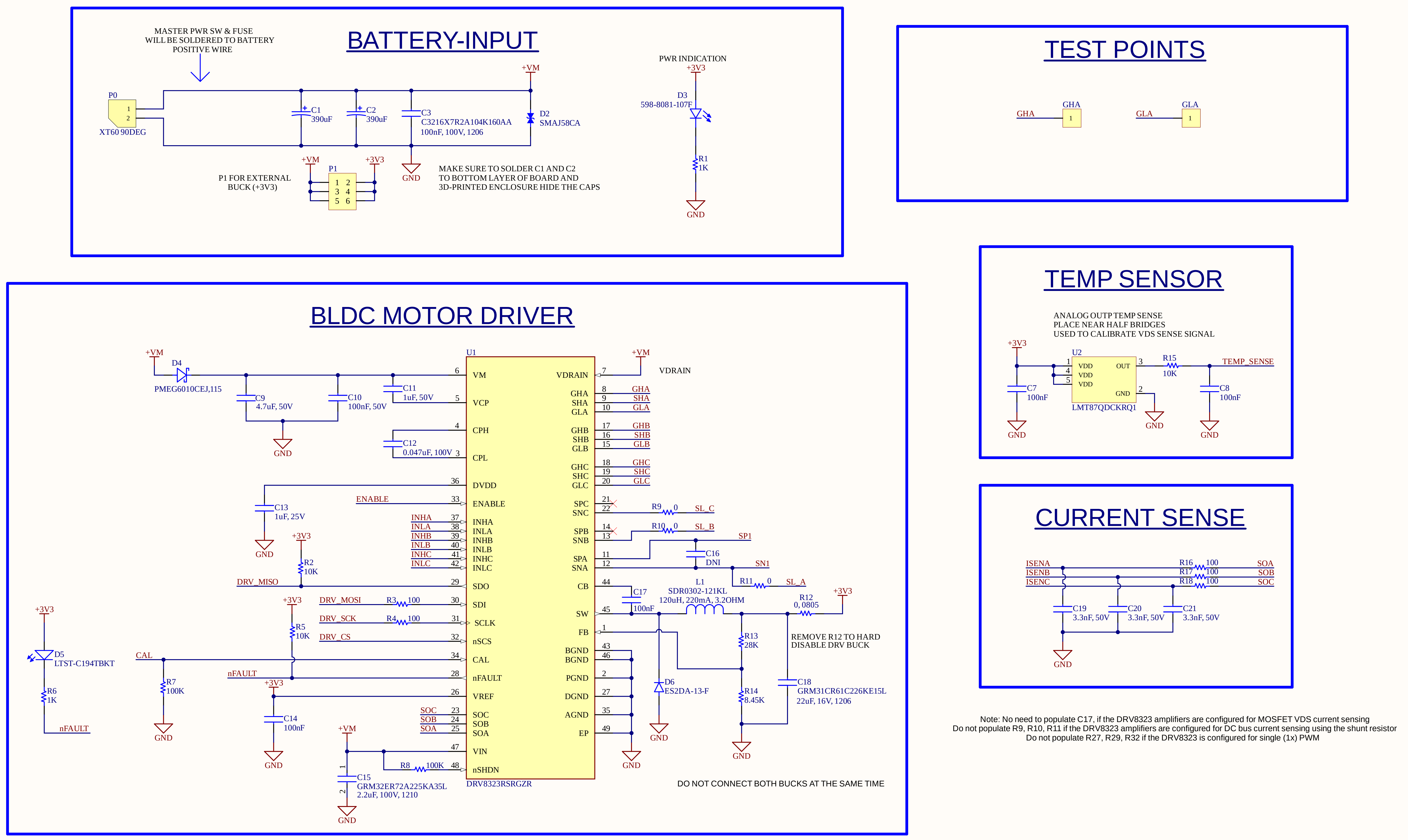
CONTRACT NO	EE400D	
DWN BY	T STOLPE	
CHECK BY	J TOLDEN	
APPV BY	J TOLDEN	
PROF	DR. JUFOUT	
MKS Rev / MKS Date	THOMAS STOLPE, JON TOLDEN, BAO PHAM, AUTUMN AGUILAR	

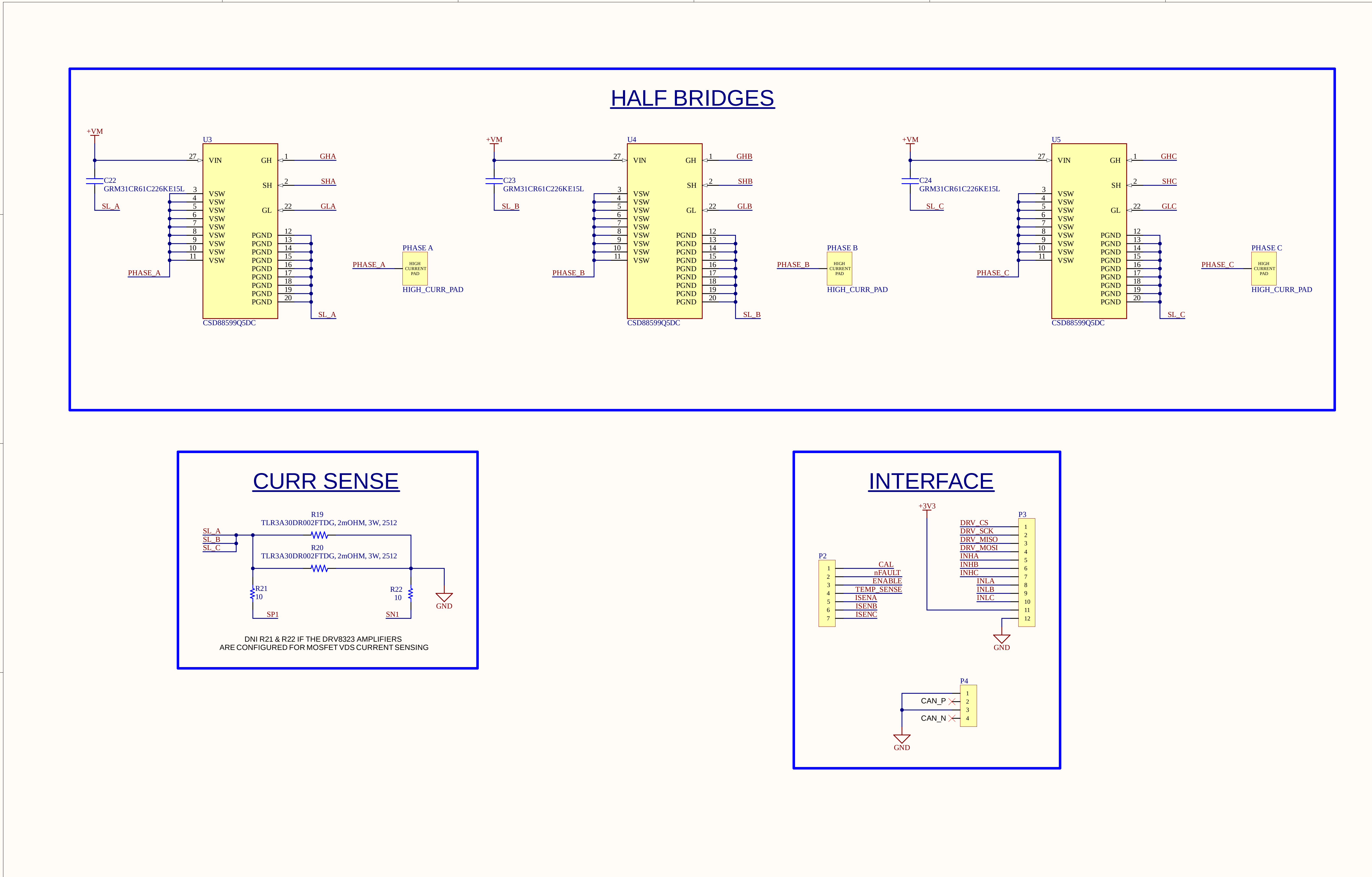
DEESC
CALIFORNIA STATE UNIVERSITY - LONG BEACH
ELECTRICAL ENGINEERING
LONG BEACH, CALIFORNIA

TITLE
MCU CONTROL FOR ESC

SIZE		DWG NO	REV
B	SPRING 2026 CSULB EE400D SENIOR DESIGN	DRV_1	-
SCALE: NONE		VIEWDRAW	SHEET 1 OF 2

BLDC MOTOR CONTROL

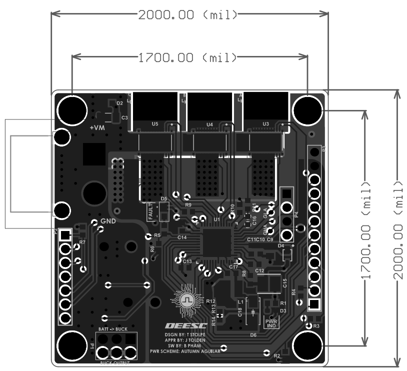




HALF BRIDGES
+VM
C22
GRM31CR61C226KE15L
SL_A
U3
VIN
GH
SH
GL
VSW
PGND
GHA
SHA
GLA
PHASE_A
SL_A
CSD88599Q5DC
PHASE A
HIGH CURRENT PAD
HIGH_CURR_PAD
C23
GRM31CR61C226KE15L
SL_B
U4
GHB
SHB
GLB
PHASE_B
CSD88599Q5DC
PHASE B
C24
GRM31CR61C226KE15L
SL_C
U5
GHC
SHC
GLC
PHASE_C
CSD88599Q5DC
PHASE C
CURR SENSE
R19
TLR3A30DR002FTDG, 2mOHM, 3W, 2512
R20
TLR3A30DR002FTDG, 2mOHM, 3W, 2512
SL_A
SL_B
SL_C
R21
10
R22
10
SP1
SN1
GND
DNI R21 & R22 IF THE DRV8323 AMPLIFIERS
ARE CONFIGURED FOR MOSFET VDS CURRENT SENSING
INTERFACE
+3V3
P2
CAL
nFAULT
ENABLE
TEMP_SENSE
ISENA
ISENB
ISENC
P3
DRV_CS
DRV_SCK
DRV_MISO
DRV_MOSI
INHA
INHB
INHC
INLA
INLB
INLC
GND
P4
CAN_P
CAN_N
GND

Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.40mil	3.5	
1	LAYER 1		1.40mil		
	Dielectric 2	PP-006	2.80mil	4.1	
2	LAYER 2	CF-004	1.38mil		
	Dielectric 1	FR-4	12.60mil	4.8	
3	LAYER 3	CF-004	1.38mil		
	Dielectric 3	PP-006	2.80mil	4.1	
4	LAYER 4		1.40mil		
	Bottom Solder	Solder Resist	0.40mil	3.5	
	Bottom Overlay				

LAYER 1: SIGNAL
LAYER 2: GND
LAYER 3: +3.3 V
LAYER 4: SIGNAL



COMPONENTS MARKED 'DNI' SHOULD NOT BE POPULATED.
ASSEMBLY VARIANT: [No Variations]

1000.00mil

DESIGN INFORMATION

MIN. TRACK WIDTH: 10 MIL
MIN. CLEARANCE: 6 MIL
MIN. VIA PAD SIZE: 20 MIL
MINIMUM ANNULAR RING 0.05mm (2MIL) EXTERNAL
PER IPC-D-275 CLASS 2 LEVEL C
REGISTRATION TOLERANCES: METAL +/- 5 MIL, HOLES +/- 3 MIL
HOLE SIZE TOLERANCE (UNLESS OTHERWISE SPECIFIED): +/- 3 MIL

MATERIAL:
[ ] FR-408 [X] FR-4 High Tg [ ] OTHER
THICKNESS: [X] 62 MIL (1.6mm) +/-10% [ ] OTHER
TOLERANCE: [X] ANSI IPC-6012 TYPE 3 CLASS 2
[ ] OTHER +/-
BOW & TWIST: [X] ANSI IPC-6012 TYPE 3 CLASS 2
[ ] OTHER +/-

DRILLING:
REFERENCE: [X] AS SHOWN [X] NC_DRILL FILES
PTH COPPER THICKNESS: [X] 20-30 um [ ] OTHER

BOARD FINISH:
SILKSCREEN: [X] TOP [X] BOTTOM
SILKSCREEN COLOR: [X] WHITE [ ] OTHER
SOLDER RESIST COLOR: [ ] GREEN [X] OTHER RED
[X] MATTE [ ] SEMI-GLOSS

SURFACE FINISH: [X] IMMERSION GOLD (ENIG) [ ] ENEPIG
[ ] IMM. TIN/SILVER OR EQUIV [ ] OTHER

ARRAY/PANEL: [ ] CUT AND TRIM PER M1 BOARD OUTLINE
[ ] N.C. ROUTE [X] V. SCORE

CERTIFICATION: MATERIALS AND WORKMANSHIP FOR ALL PCBS TO MEET OR EXCEED THE REQUIREMENTS OF:
[X] ANSI IPC-A-600F CLASS -> [ ] 1 [X] 2 [ ] 3
[X] RoHS [ ] OTHER PER ORDER

ALL BOARDS MUST MEET OR EXCEED EE400D REQUIREMENTS.

ADDITIONAL REQUIREMENTS:
MICROSECTION: [ ] YES
BARE BOARD ELEC. TEST: [ ] NONE [X] REQUIRED [ ] PER ORDER

DEESC

TEAM LEAD: JON TOLDEN
THOMAS STOLPE
BAO PHAM
AUTUMN AGUILAR

PROJECT TITLE:
ESC PCB FOR BLDC MOTOR

DESIGNED FOR:
CSULB EE400D SENIOR PROJECT

FILE NAME:
ESC_EE400D_PCB.PcbDoc

PROJECT MANAGER: JON TOLDEN
LAYOUT BY: THOMAS STOLPE
SCALE: 0.72
ALTIUM DESIGNER VERSION: 26.8.1.31

PCB VIEWED FROM TOP SIDE
SOFTWARE BY: BAO PHAM
POWER SCHEME BY: AUTUMN AGUILAR
GENERATED : 8/7/2026 8:51:05 PM